

Public Debt and Changing Inflation Targets[†]

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What are the effects of a higher central bank inflation target on the burden of real public debt? Several recent proposals have suggested that even a moderate increase in the inflation target can have a pronounced effect on real public debt. We consider this question in a New Keynesian model with a maturity structure of public debt and an imperfectly observed inflation target. We find that moderate changes in the inflation target only have significant effects on real public debt if they are essentially permanent. Moreover, the additional benefits of not communicating a change in the inflation target are minor. (JEL E12, E31, E52, H63)

Large increases in government deficits during the economic crisis of 2008 and 2009 initiated a debate on whether the real value of public debt should be reduced by at least temporarily raising inflation. For example, it has been claimed that increasing the US inflation rate by 4 percentage points for a couple of years would significantly help the process of deleveraging public (as well as private) sector balance sheets. It would thus also reduce the need for possibly contractionary fiscal consolidations.¹ The same considerations potentially apply to the Eurozone, which suffers from even larger debt problems than the United States. Underlying the debate is the presumption that inflation is always effective in reducing public debt, and that policy makers can always generate a desired level of inflation. This may be difficult, particularly at times when nominal interest rates are at the zero lower bound.

Two factors determine how effective raising the targeted inflation rate is in reducing real public debt: long-term inflation expectations and the maturity structure of debt. On the one hand, inflation expectations affect current inflation through

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¹ Most notably, Kenneth Rogoff, in Project Syndicate, has at the end of 2008 and 2010 proposed to allow some 6 to 7 percent inflation for a short time. See Miller (2009) for more details. See also Rajan (2011), who discusses a number of arguments for and against inflating away debt, raising informally some of the issues made explicit here.

forward-looking price setting, and they affect long-term nominal interest rates, and thus the cost of servicing debt, through the pricing of newly issued debt. How inflation expectations evolve depends crucially on the perception of the central bank's inflation target and its credibility. On the other hand, the maturity structure of debt determines how much of the outstanding debt can be inflated away even for inflation that is anticipated, for a potentially large fraction of debt remains priced under the lower long-term inflation expectations that prevailed before such a change in policy.

This paper quantitatively analyses how the interaction of inflation expectations and debt maturity shapes the evolution of real public debt after an increase in a central bank's target inflation rate. To this end, we set up a New Keynesian business cycle model with two nonstandard features. First, the inflation target in the interest rate rule is stochastic, and we allow for the possibility that agents have imperfect information in that the current target has to be inferred from the monetary authority's behavior. Technically, we assume that agents face a signal extraction problem as to whether observed changes in monetary policy are due to transitory shocks or persistent changes in the inflation target. The speed at which economic agents revise their beliefs of the target rate can be interpreted as a measure of the credibility of a previously communicated inflation target.² A perceived low inflation target may keep inflation expectations, and thus nominal interest rates, on newly issued debt low, potentially raising the amount of debt that can be inflated away in subsequent periods.

Second, to model the maturity structure of long-term debt, we introduce perpetuities with stochastically arriving call date, which implies that a given fraction of debt matures each period. This type of bond offers a simple means to calibrate a realistic average maturity of public debt and to track the average interest rate on outstanding debt. It thus reveals countervailing effects of changes in the inflation target not apparent in standard models with only short-term debt. In fact, in OECD countries in 2013, the average maturity of debt ranged between about four to seven years, so significant fractions of real debt are susceptible to increases in inflation even when inflation expectations, and thus interest rates demanded for newly issued debt, have adjusted. For example, an average maturity of 7 years implies that per year only 20 percent of debt is rolled over. In all other respects, the model is a standard dynamic stochastic general equilibrium model with monopolistic competition and sticky prices, and with policy described by a Taylor-type interest rate rule and a fiscal tax rule.

For the simulation of the model, we define the steady-state debt-to-output ratio as the level prevailing before the crisis and implement a "debt shock" that brings that ratio to the level observed in 2013. Then we trace the evolution of the real value of public debt after a contemporaneous increase of the inflation target in the interest rate rule. For concreteness, we calibrate our benchmark economy to the United States with a current average maturity of debt of about 5.5 years, and set the debt shock to 65 percent of steady-state debt, the actual increase in US net government debt between 2008 and 2013. We then compare how different outcomes depend on

² As in Erceg and Levin (2003), this problem is solved by means of the Kalman Filter. For an early application of this conceptualization of credibility, see Cukierman and Meltzer (1986).

the main parameters of interest and on the way expectations are formed. In particular, we vary the size and persistence of the target change from its steady-state value and the average maturity of debt, and compare the cases of full and imperfect information about the change in the inflation target.

We find that a significant fraction of real public debt can be inflated away only if the change in the inflation target is very persistent.³ For the US calibration, a persistent 4 percentage point increase in the target erodes after 10 years, about a third of the additional debt accrued during the crisis. A temporary change, similar to what, for example, Rogoff (2008, 2010) suggested, has much smaller effects, of about 10 percent of debt after 10 years. The reason for this difference is that only when the higher inflation is long-lasting can the fact be exploited that a large fraction of debt remains priced at low interest rates, as the fraction of debt maturing and rolled over is relatively small. With only short-term debt, as standard in New Keynesian models, the effect of inflation on debt is of course negligible.⁴ Conversely, a higher maturity increases the effect on real debt. For example, with an average maturity of roughly 14.5 years as in the United Kingdom in 2013, the additional real debt induced by the crisis that can be reduced after 10 years is about 42 percent for a 4 percentage point increase in the inflation target. Thus, a country with a higher average maturity could be regarded as facing a larger temptation to increase inflation.

What if the central bank does not communicate a change in the target? This is the case of imperfect information described above. Then, if the credibility of a previously established target is high, inflation expectations would continue to stay low in spite of actually rising inflation. Since investors are not compensated by higher interest rates even on rolled-over long-term debt, the real debt burden of the government may fall more strongly than under full information. However, it turns out in the simulations that for realistic maturities and a persistent change in the target, such a communication strategy has a very low additional effect because the fraction of incorrectly priced new debt is small. Only at very low, but implausible, average maturities, does the mispricing of debt under imperfect information play a relatively larger role, since more debt is rolled over and therefore insufficiently compensating for the higher inflation that follows.

By the end of 2013, nominal short-term interest rates in the United States, the Euro Area, and other developed economies still were essentially at the zero lower bound. This has consequences for the ability of reducing real public debt through inflation because it may not be possible to raise inflation in the first place. We therefore simulate a scenario where the zero lower bound is binding after a contractionary demand shock. Absent changes in the inflation target, output falls and inflation turns negative, adding substantially to the increase in real public debt. Now if a simultaneous increase in the inflation target is persistent and perceived with full information, the deflation is smaller and an exit from the zero lower bound takes

³Persistence is measured by the autoregressive parameter of the process describing the evolution of the inflation target. We call a persistent change one with an AR(1) coefficient of 0.99 and temporary change one with a coefficient of 0.85. The half-lives are about 17 years and 1 year, respectively.

⁴As will become clear in the analysis, our findings present an upper bound since the maturity of debt is assumed not to change after the target change. If the maturity structure were to change, and debt immediately repriced, the "gain" from raising inflation would be even lower.

place earlier, so that real debt also rises by much less. These effects are driven by the correctly perceived inflation target, and consequently by expectations of higher inflation in the future. By contrast, when announcements about the inflation target are not credible, there is no such forward guidance, and inflation can not possibly be raised until after the economy has exited from the zero lower bound. Thus the ability to reduce real public debt by raising inflation depends on the ability of monetary policy to commit to a higher inflation target in the future, relying on the credibility of its announcements.

Two other studies explicitly analyze the possibility of the US government to inflate away its real debt. Aizenman and Marion (2011) document the evolution of debt, inflation, and the maturity of debt since World War II. In a simple partial equilibrium model with a fixed interest rate they show that the incentives to inflate to reduce debt as of 2011 are large, arising from the capital loss imposed on foreigners. By incorporating endogenous interest rates and forward-looking expectations in a full general equilibrium setup, we show how the effects of raising inflation depend on the process that the inflation target is expected to follow, and that the benefits of inflation are potentially rather low. Hilscher, Raviv, and Reis (2014) take financial market data to extract a probability distribution of likely inflation outcomes, as perceived by market participants. Taking account of the maturity structure and holding structure of US government debt, they determine how much debt held by the public is likely to be inflated away under the highest inflation rates expected by the market. They conclude that the magnitude of debt debasement is low, unless either financial repression comes into place, or if inflation follows a path currently regarded as unlikely. In this paper, we evaluate the consequences on public debt of such a counterfactual scenario. Also related are Hall and Sargent (2011) and Giannitsarou and Scott (2008). In a paper about the measurement of interest paid on government debt, Hall and Sargent show that under their measures, the fraction of US real debt inflated away after World War II is about 15 percent, which is still significant, but lower than previously estimated. Furthermore, they emphasize, in line with Giannitsarou and Scott, that high real GDP growth made the largest contribution to real debt reduction, and not inflation.⁵

Leeper and Zhou (2013) analyze the role of government debt maturity for the stabilization of output and inflation, and show that for higher debt optimal policy shifts toward using more inflation rather than distortionary taxation. This suggests that the currently observed increase in debt is likely to lead to higher inflation, but for other reasons than those analyzed in this paper. Also related are Davig, Leeper, and Walker (2011) who show how increases in public debt may endogenously lead to a switch in the monetary policy regime when debt reaches a “fiscal limit.” In other words, high debt may trigger the central bank to be passive while fiscal policy actively determines the price level, in the terminology of Leeper (1991). Once a fiscal limit is reached, monetary policy switches stochastically to a passive stance, and inflation serves to bring the public debt back to a sustainable level. In our paper, we focus on the role of debt maturity and the public’s beliefs about the inflation target

⁵ See Persson, Persson, and Svensson (1996) for an attempt to address this issue for the Swedish case.

in a regime where monetary policy remains active and fiscal policy passive. A recent study by Bianchi (2013) introduces beliefs about the likelihood of different regimes to interpret past inflationary periods, such as the 1970s, which he finds to be one of low credibility.

The paper now proceeds to the following section where we give a brief overview of the current fiscal situation of advanced G7 economies, including details on the maturity structure of debt. Section II develops the model, and introduces the long-term bond, or callable perpetuity, designed to approximate a realistic maturity structure of government debt, and specifies the signal-extraction problem regarding the long-term inflation target of the central bank. In Section III, the analysis is presented, followed by a discussion focusing on debt maturity and the persistence of the inflation target. In Section IV, we address the question of how sensitive real public debt is to changing inflation targets, and the role of the zero lower bound on nominal interest rates. Finally, Section V concludes.

I. Public Debt and Maturity in Advanced Economies

Since the onset of the economic and financial crisis in 2008, advanced economies have experienced rising levels of public debt due to financial sector rescue packages, fiscal stimuli, and falling tax revenues. On average, net debt in G7 countries has increased from 52 percent of GDP in 2006 to 86 percent in 2013, as reported by the IMF. Debt is projected to increase further in the coming years. The corresponding gross debt percentages are 83 and 119 percent, respectively.⁶

The average maturity of public debt varies across countries, but is largely in the range of four to seven years. A notable outlier is the United Kingdom with an average maturity of 14.4 years, which, however, includes about 30 percent indexed debt. Nonetheless, a large fraction of the real value of UK debt remains to be potentially affected by higher inflation, before higher interest rates on rolled-over debt would increase the government's interest expenses. In the wake of the crisis, countries have adjusted the maturity of newly issued debt, in order to manage their future repayment obligations. For example, the United States had relatively low average maturity in 2009, due to large issuance of short-term paper to finance crisis-related expenses, but aims at increasing its average maturity.⁷ Overall, there is typically some variation over time in the average maturity of public debt, depending on the

⁶Gross debt comprises government debt held by the government, such as the social security trust fund in the United States. Net debt excludes such government assets and thus a better measures of what the government owes the private sector. In the United States this is reported as *debt held by the public*, which also includes holdings of treasury bonds by the Federal Reserve System. The asset purchases of the Federal Reserve System have increased those holdings, which stood at 1.6 billion dollars by the end of 2012 (see Hilscher, Raviv, and Reis 2014). The last four columns of Table 1 show net government debt in percent of GDP and the percentage change of net debt between 2008 and 2013, as well as the same measures relating to gross debt.

⁷As of April 2009. Taken from US Department of the Treasury (2009) Report to the Secretary of the Treasury from the Treasury Borrowing Advisory Committee of the Securities Industry and Financial Markets Association, April 29, 2009. See also the Report from August 5, 2009 which reports that "... the average maturity of issuance now exceeds the average maturity of marketable debt outstanding. This suggests that the decline in the average maturity of debt outstanding that we have witnessed over the past seven years—from a high of approximately 70 months in 2000 to a low of approximately 50 months earlier this year should be arrested and begin to slowly lengthen going forward."

TABLE 1—PUBLIC DEBT AND MATURITY STRUCTURES, 2013

	Avg. maturity in years	% of debt maturing	Net debt % of GDP	% change from 2008	Gross debt % of GDP	% change from 2008
United States	5.3	18.6	89.0	+64.8	108.1	+43.2
France	6.8	13.4	86.5	+38.8	92.7	+35.9
Germany	6.4	7.9	56.2	+12.2	80.4	+20.4
Italy	6.5	25.3	105.8	+19.1	130.6	+23.1
Japan	6.3	49.2	143.4	+14.1	245.4	+27.9
United Kingdom	14.4	6.1	86.1	+79.0	93.6	+79.3
Canada	5.1	13.3	35.9	+60.3	87.0	+22.0
Simple average	7.26	19.1	86.1	+41.2	119.7	+36.0

Source: G7 Advanced Economies. IMF Fiscal Monitor April 2013, tables 2 and 6, statistical table 12a. Debt levels for 2013 are IMF projections.

history of issuance. There is also some variation across countries in the fraction of debt maturing, which results from differences in the distribution of maturities. For example, while Japanese debt has an average maturity of 6.3 years, rather close to the 5.3 years in the United States, the fraction of debt maturing in 2013 is 49.2 percent, which is much higher than the 18.6 percent for the United States. This drives up the average in Table 1 to 19.1 percent. To achieve the high average maturity of debt, there must be a larger fraction of relatively long-term debt outstanding for Japan. For the United Kingdom, the relative numbers are more in line with intuition, in that a higher average maturity of debt than in the United States implies a lower fraction of debt maturing.

II. A Model of Long-Term Debt and Inflation

The model is a standard New Keynesian framework, with the addition of long-term bonds with stochastic maturity and of learning about the inflation target. The central bank follows a Taylor rule which sets the short-term nominal interest rate as a function of the inflation rate gap and the output gap; there is a one-period government bond priced at that interest rate, in addition to the newly introduced stochastic bond. Firms are monopolistic competitors selling differentiated products at prices that are allowed to adjust in a stochastic fashion as in Calvo (1983).⁸ Consumers maximize lifetime utility from consumption, labor input, and real money holdings. Government budget dynamics are determined by a fiscal rule.

A. The Maturity Structure of Public Debt

The central element of the model is an approximation of the maturity structure of public debt in terms of a stochastic, long-term, bond. Each period, an individual bond of this type pays the interest determined when the bond was issued and, with a given probability, matures. In this case it also pays back the principal. Technically,

⁸Our results are robust to the inclusion of sticky wages à la Erceg, Henderson, and Levin (2000), thus we stick to the simplest model.

the bond could be regarded as a callable perpetuity, but one where the call date is stochastic and independent across bonds. Since the government issues a large number of these bonds each period, the fraction of bonds maturing is identical to the call probability. Private agents are assumed to hold the representative portfolio of the bonds.⁹

With probability α the stochastic bond matures, and with probability $1 - \alpha$ it survives into the next period. Denote the total value of the stock of long-term bonds with B_t^L . The total stock of long-term bonds then evolves as

$$(1) \quad B_t^L = (1 - \alpha)B_{t-1}^L + B_t^{L,n},$$

where $B_t^{L,n}$ denotes the value of newly issued bonds, while $(1 - \alpha)B_{t-1}^L$ is the value of bonds not maturing. Every period, the government is assumed to issue new debt, to replenish the depleted debt, and to reduce or increase the total amount of outstanding debt. There is always a stock of bonds that was not redeemed, and all ages of bonds are present in the market.

Let the interest rate of bonds newly issued in period t be given by $i_t^{L,n}$, and the average interest rate of all current and previously issued stochastic bonds by i_t^L . Then the latter is given by

$$i_t^L = \frac{B_t^{L,n}}{B_t^L} i_t^{L,n} + (1 - \alpha) \frac{B_{t-1}^{L,n}}{B_t^L} i_{t-1}^{L,n} + (1 - \alpha)^2 \frac{B_{t-2}^{L,n}}{B_t^L} i_{t-2}^{L,n} + \dots$$

The weights on the interest rates of previously issued bonds depend on the fraction of those bonds that has survived until date t and the value of these bonds relative to that of the current stock of long-term debt. Thus, the average interest rate on outstanding long-term debt can conveniently be tracked in recursive form:

$$(2) \quad i_t^L B_t^L = (1 - \alpha) i_{t-1}^L B_{t-1}^L + i_t^{L,n} B_t^{L,n}.$$

The interest rate $i_t^{L,n}$ is determined according to an appropriate arbitrage condition between the one-period and the stochastic bonds, derived below from the households first-order conditions.

The parameter α determines not only the fraction of debt maturing each period, but also the average maturity of outstanding debt, $1/\alpha$. We calibrate the latter to match the actual average maturity of debt in the data. Below we show that this choice of α also gives a good approximation of the observed fractions of debt maturing, which is important for the effects of changes in inflation on the real value of outstanding debt.

⁹We exclude the possibility of explicit government default other than implicitly by inflation. See Hatchondo and Martinez (2009) or Arellano and Ramanarayanan (2012) for recent examples. Also, we do not explore inflation risk premia and term structure implications of our model. On this, see, for example, Rudebusch and Swanson (2008). These authors use, as in Woodford (2001), an assumption on declining payment streams on consols, which in the aggregate shows some similarities with the bond structure developed here.

B. Households

The representative household is assumed to maximize the present value of utility

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma_c}}{1-\sigma_c} + \chi \frac{(M_t/P_t)^{1-\sigma_m}}{1-\sigma_m} - \varphi \frac{N_t^{1+\phi}}{1+\phi} \right),$$

with C_t consumption, M_t/P_t real money balances, and β the discount factor, σ_c the inverse of the inter-temporal elasticity of substitution (and the inverse of risk aversion), σ_m governs the interest elasticity of money demand, and χ a utility weight. Labor services provided enter negatively, with ϕ the inverse of the Frisch elasticity of labor supply, and φ scaling labor disutility. The consumption good is an aggregate of a continuum of differentiated products $C_t(z)$, and given by the function $C_t = \left(\int_0^{\infty} C_t(z)^{\frac{\epsilon-1}{\epsilon}} dz \right)^{\frac{\epsilon}{\epsilon-1}}$, with $\epsilon > 1$ a constant elasticity of substitution. The individual goods are supplied by a continuum of monopolistically competitive firms at price $P_t(z)$ for each firm z .

Maximization takes place subject to the evolution of the interest rate on the portfolio of bonds (2), and the budget constraint

$$\begin{aligned} \frac{B_t}{P_t} + \frac{B_t^{L,n}}{P_t} + \frac{M_t}{P_t} + C_t &= (1 + i_{t-1}) \frac{B_{t-1}}{P_t} + (\alpha + i_{t-1}^L) \frac{B_{t-1}^L}{P_t} + \frac{M_{t-1}}{P_t} \\ &+ (1 - \tau_t) \frac{W_t}{P_t} N_t + \int_0^1 \frac{\Pi_t(z)}{P_t} dz, \end{aligned}$$

where W_t/P_t is the real wage and τ_t is a proportional tax rate on labor income. $\Pi_t(z)$ is nominal income from dividends of monopolistically competitive intermediate firms—indexed z —owned by households. A one-period bond issued in period t is denoted by B_t and pays interest i_t in the following period. It is in zero net supply in equilibrium. In contrast, only a fraction α of long-term bonds B_{t-1}^L is redeemed in period t , and a quantity $B_t^{L,n}$ of bonds are newly issued.

Combining equation (1) with equation (2) and the budget constraint, the representative household maximizes its intertemporal utility with respect to C_t , B_t , B_t^L , i_t^L , M_t , N_t , and $C_t(z)$. Note at this point that, from the perspective of an individual household, while the market-determined long-term interest rate $i_t^{L,n}$ is taken as given, the average interest rate i_t^L depends on the composition of newly issued relative to outstanding bonds that the households chooses to hold. This must be taken into account when solving the household's optimization problem.

Consumption smoothing and the holdings of the two types of bonds are guided by the familiar Euler equation for short-term bonds,

$$(3) \quad 1 = E_t \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{P_t}{P_{t+1}} [1 + i_t],$$

and a similar Euler equation for long-term bonds,

$$(4) \quad 1 = E_t \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{P_t}{P_{t+1}} [1 + i_t^{L,n} - \mu_{t+1}(1 - \alpha) \Delta i_{t+1}^{L,n}],$$

where λ_t is the marginal utility of wealth, which must be equal to the marginal utility of consumption

$$(5) \quad \lambda_t = C_t^{-\sigma_c}.$$

The second Euler condition deserves further comment. It relates the nominal stochastic discount factor $\beta(\lambda_{t+1}/\lambda_t)P_t/P_{t+1}$ to the interest rate on newly issued long-term debt, $i_t^{L,n}$, corrected for its expected change, $\Delta i_{t+1}^{L,n} = i_{t+1}^{L,n} - i_t^{L,n}$. The change is valued by using the Lagrange multiplier on (2), which is the price of the stochastic bond, and which must follow

$$(6) \quad \mu_t = E_t \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{P_t}{P_{t+1}} [1 + (1 - \alpha)\mu_{t+1}],$$

for an expected quarterly payout of one.¹⁰ Thus, $-\mu_{t+1} \Delta i_{t+1}^{L,n}$ in (4) is the capital loss (or gain) in period $t + 1$ incurred from a rise (or fall) in the long-term interest rate. The intuition of (4) is that an expected capital loss on long-term bonds reduces the incentives to invest in long-term bonds today, and thus requires the long-term interest rate today, $i_t^{L,n}$, to be higher in order to ensure that agents are indifferent to investing in short-term bonds.

The remaining optimality conditions are the familiar conditions for money demand,

$$(7) \quad \frac{M_t}{P_t} = \left[\chi C_t^{\sigma_c} \frac{1 + i_t}{i_t} \right]^{1/\sigma_m},$$

labor supply,

$$(8) \quad \varphi N_t^\phi = C_t^{-\sigma_c} (1 - \tau_t) \frac{W_t}{P_t},$$

and the demand for differentiated products, $C_t(z) = \left(\frac{P_t(z)}{P_t} \right)^{-\epsilon} C_t$, where the price level is defined as the cost of the minimum-expenditure combination of the $C_t(z)$ to obtain a given value of C_t , $P_t \equiv \left(\int_0^\infty P_t(z)^{1-\epsilon} dz \right)^{1/(1-\epsilon)}$.

¹⁰In steady state, the price is $\mu = \frac{1}{i + \alpha}$, which gives the familiar price equations for the one-period bond when $\alpha = 1$ and for a consol when $\alpha = 0$.

C. Firms

Firms are monopolistic competitors each facing iso-elastic demand for their differentiated products derived above, and demand labor to produce. Production is linear in labor, $Y_t(z) = AN_t(z)$, where A is the aggregate productivity level. Prices are sticky in that each period, following Calvo (1983), only a fraction $(1 - \theta)$ of firms is able to optimally adjust prices. If a firm cannot re-optimize its price, the nominal price evolves according to the indexation rule $P_t(z) = \pi_t^* P_{t-1}(z)$, where π_t^* is the actual inflation target. Under imperfect information, as introduced later, π_t^* will have to be replaced by the perceived inflation target. Thus, $\pi_t = P_t/P_{t-1}$ is the gross aggregate inflation rate. The inclusion of the actual or perceived inflation target in the indexing rule is crucial for the issue at hand because we deal with potentially permanent changes in inflation and want to ensure that long-run monetary neutrality holds.¹¹

Taking into account that it might not be able to set its price optimally in a near future, a firm z chooses the optimal price, $P_t^*(z)$, by maximizing intertemporal profits subject to the demand it faces and taking into account the indexing rule. The first-order condition for this program is

$$\frac{P_t^*}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{Z_{1,t}}{Z_{2,t}},$$

where

$$(9) \quad Z_{1,t} = \lambda_t mc_t C_t + \theta \beta E_t \left[\left(\frac{\pi_{t+1}}{\pi_{t+1}^*} \right)^{-\epsilon} Z_{1,t+1} \right]$$

and

$$(10) \quad Z_{2,t} = \lambda_t C_t + \theta \beta E_t \left[\left(\frac{\pi_{t+1}}{\pi_{t+1}^*} \right)^{1-\epsilon} Z_{2,t+1} \right],$$

which is the same for all firms that can adjust their price in period t . Real marginal costs are given by $mc_t = (W_t/P_t)/A$ and λ_t is the marginal utility of consumption, which appears by the assumption of perfect capital markets. The aggregate price index can be shown to evolve according to

$$(11) \quad 1 = \theta \pi_t^{*(1-\epsilon)} \pi_t^{-(1-\epsilon)} + (1 - \theta) \left(\frac{\epsilon}{\epsilon - 1} \frac{Z_{1,t}}{Z_{2,t}} \right)^{1-\epsilon}.$$

¹¹We also experimented the more general indexation scheme $P_t(z) = \tilde{\pi}_t P_{t-1}(z)$, where we allowed $\tilde{\pi}_t = \pi_{t-1}^\xi \pi_t^{*(1-\xi)}$ to depend on both lagged actual inflation and the actual (or perceived) inflation target π_t^* . We found that our main results are barely affected by this modification.

D. The Fiscal and Monetary Authorities

The fiscal authority follows a rule that adjusts the tax rate depending on the deviation of real debt from a long-run level of real debt, assumed as given. The tax rule is given by

$$(12) \quad \tau_t - \tau = \rho_\tau(\tau_{t-1} - \tau) + \phi_\tau \hat{b}_t^L,$$

where τ is the steady-state tax rate and $\hat{b}_t^L$ is the percent deviation of total real long-term debt, $b_t^L = B_t^L/P_t$, from its long-run steady-state level. The tax smoothing parameter ρ_τ prevents excessive jumps in the tax rate, and ϕ_τ determines the responsiveness of the tax rate to variations in real debt. Since the one-period bond $b_t = B_t/P_t$ is in zero net supply, it is only relevant for the pricing of assets via the short-term policy interest rate i_t .

Aggregate public debt then evolves according to the consolidated budget constraint of the public sector, written here in real terms as

$$(13) \quad \tau_t w_t N_t + m_t - \frac{m_{t-1}}{\pi_t} + b_t^{L,n} = g + (\alpha + i_{t-1}^L) \frac{b_{t-1}^L}{\pi_t}.$$

Government revenue consists of tax revenue $\tau_t w_t N_t$, seignorage revenue, where $m_t = M_t/P_t$, and newly issued debt, $b_t^{L,n}$, while expenditure consists of (exogenous) real government spending g , redeemed bonds αb_{t-1}^L , and real interest paid on bonds $(i_{t-1}^L b_{t-1}^L)/\pi_t$. Rewriting the evolution of long-term debt (1) in real terms

$$b_t^L = (1 - \alpha) \frac{b_{t-1}^L}{\pi_t} + b_t^{L,n},$$

shows how much of the last period's outstanding debt, b_{t-1}^L , after accounting for the possible effects of inflation, π_t , is carried over to the current period.

The monetary authority follows an interest rate rule given by

$$(14) \quad i_t = \rho_i i_{t-1} + (1 - \rho_i) \left[i + \hat{\pi}_t^* + \phi_\pi (\hat{\pi}_t - \hat{\pi}_t^*) + \phi_y (\hat{Y}_t - \hat{Y}_t^n) \right] + \eta_t,$$

with i the steady-state value of the nominal interest rate, $\hat{\pi}_t^*$ the time-varying inflation target, $\hat{Y}_t$ is actual output, and $\hat{Y}_t^n$ is the natural rate of output being defined as the level of output that would prevail under fully flexible prices, all expressed as deviations from steady state.¹²

Hence, $i + \hat{\pi}_t^*$ is the variation of the nominal rate that is governed by changes in the inflation target. The policy interest rate adjusts with inertia, as given by ρ_i . The

¹²As our model features several endogenous state variables, we assume, following Woodford (2002), that the natural output is determined conditionally on the sticky price model's debt, average interest rate and money stock levels. For an extensive and critical discussion of this concept we refer the reader to the Appendix of Neiss and Nelson (2001).

interest rate rule is additionally subject to a monetary policy white noise disturbance η_t with variance σ^2 . The policy rule as perceived by agents under imperfect information is introduced later.

The percentage deviation of the inflation target from steady state is assumed to evolve according to the AR(1) process:

$$\hat{\pi}_t^* = \rho_\pi \hat{\pi}_{t-1}^* + \eta_t^\pi,$$

with η_t^π a white noise process with variance σ_π^2 .

The persistence parameter ρ_π is between zero and one so that variations of the target are potentially very persistent. The shocks are independently and identically distributed normal. To reflect a high degree of credibility of an existing target in the imperfect information scenario, the variance of η_t^π is assumed substantially lower than that of η_t .

E. Market Clearing and Equilibrium

Aggregate demand is given by total private and government consumption:

$$(15) \quad Y_t = C_t + g,$$

and the market clearing condition on goods market is given by

$$(16) \quad \Delta_{p,t} Y_t = AN_t,$$

where $N_t = \int_0^1 N_t(z) dz$ is aggregate labor input and the term $\Delta_{p,t} = \int_0^1 \left(\frac{P_t(z)}{P_t} \right)^{-\epsilon} dz$ measures the price dispersion arising from staggered price setting. Similar to the aggregate price index, the price distortion has a law of motion that can be shown to be

$$(17) \quad \Delta_{p,t} = \theta \Delta_{p,t-1} \left(\frac{\pi_t}{\pi_t^*} \right)^\epsilon + (1 - \theta) \left(\frac{\epsilon}{\epsilon - 1} \frac{\mathcal{Z}_{1,t}}{\mathcal{Z}_{2,t}} \right)^{-\epsilon}.$$

The competitive equilibrium of our model is a set of stationary processes $B_t^L, B_t^{L,n}, C_t, \Delta_{p,t}, i_t, i_t^L, i_t^{L,n}, \lambda_t, M_t, \mu_t, N_t, \pi_t, \tau_t, W_t, Y_t, \mathcal{Z}_{1,t}, \mathcal{Z}_{2,t}$, satisfying the relations (1) to (17), given the exogenous stochastic processes η_t, η_t^π , and Y_t^* , and the initial conditions $B_{-1}, B_{-1}^L, i_{-1}, i_{-1}^L, \Delta_{p,-1}$, and π_{-1} .

F. Imperfect Information and Credibility

We allow for the possibility that private agents do not have perfect knowledge of the central bank's objectives. That is, agents cannot distinguish movements in the inflation target from movements in the monetary policy shock, but only receive a signal on an aggregate monetary policy shock, defined as

$$(18) \quad \varepsilon_t^\pi \equiv (1 - \rho_i)(1 - \phi_\pi) \hat{\pi}_t^* + \eta_t.$$

The signal extraction problem entails backing out the two components $\hat{\pi}_t^*$ and η_t in the Taylor rule (14). Formally, given their knowledge about the driving process of the shocks and of the standard deviation of the inflation target and policy shock, agents use a simple Kalman filter to extract the optimal estimates of the two unobserved components of ε_t^π .¹³

The optimal estimate of the inflation target evolves according to

$$(19) \quad \tilde{E}_t \hat{\pi}_t^* = \tilde{E}_{t-1} \hat{\pi}_t^* + \frac{k}{\rho_\pi} (\varepsilon_t^\pi - \tilde{E}_{t-1} \varepsilon_t^\pi),$$

where $k = \frac{\rho_\pi(1 - \rho_i)(1 - \phi_\pi)\mathcal{P}}{((1 - \rho_i)(1 - \phi_\pi))^2\mathcal{P} + \sigma^2}$ is the Kalman gain parameter of the steady-state Kalman filter, with $\mathcal{P}$ solving $\mathcal{P}^2 + [(1 - \rho_\pi^2)\sigma^2/[(1 - \rho_i)(1 - \phi_\pi)]^2 - \sigma_\pi^2]\mathcal{P} - (\sigma_\pi\sigma/[(1 - \rho_i)(1 - \phi_\pi)])^2 = 0$. Note that a higher variance of the monetary policy shock relative to the variance of the target shock implies a lower Kalman gain. Correspondingly, the optimal estimate of the monetary policy shock is

$$(20) \quad \tilde{E}_t \eta_t = \varepsilon_t^\pi - (1 - \rho_i)(1 - \phi_\pi)\tilde{E}_{t-1} \hat{\pi}_t^*.$$

Then the optimal forecasts of the future inflation targets and monetary policy shock can be obtained:

$$\begin{bmatrix} \tilde{E}_t \hat{\pi}_{t+i}^* \\ \tilde{E}_t \eta_{t+i} \end{bmatrix} = \begin{bmatrix} \rho_\pi & 0 \\ 0 & 0 \end{bmatrix}^i \begin{bmatrix} \tilde{E}_t \hat{\pi}_t^* \\ \tilde{E}_t \eta_t \end{bmatrix}.$$

It is important to keep in mind that, if the intention of a higher inflation rate is announced and believed, expectations of future inflation are correct in the sense that agents would not be making systematic errors in predicting inflation. In other words, the signal extraction problem would be absent, as this is the full information case. The signal extraction problem is used to capture different degrees of credibility of the central bank's established inflation target. Under imperfect information, i.e., when the central bank does not announce a changed inflation target—or agents do not believe an announcement—agents repeatedly make forecast errors, since over many periods their perception of the target differs from the actual realization of the target. Slow learning about the true increasing target reflects a high credibility of a previously prevailing—low—inflation target. Changes in ε_t^π will be ascribed mainly to the transitory shocks, and inflation expectations for a longer time will remain anchored at a low level. Our analysis is thus the opposite to the case considered in Erceg and Levin (2003), who analyze slowly revised perceptions about a reduction in the Federal Reserve's inflation target, as it took place during the Volcker disinflation in the 1980s.

¹³ Examples of such kind of imperfect information mechanism can be found in Erceg and Levin (2003); Melecký, Rodríguez Palenzuela, and Söderström (2009); Fève, Matheron, and Sahuc (2010); or Darracq-Paries and Moyen (2012).

G. Parametrization and Solution Procedure

The model is parametrized at the quarterly frequency, with a discount factor of $\beta = 0.99$, which implies a steady-state annual real interest rate of about 4 percent. The intertemporal elasticity of substitution is governed by $\sigma_c = 1.5$ following the estimates in Smets and Wouters (2007), and the disutility of labor is determined via $\varphi = 2$ in line with Domeij and Flodén (2006). We set the money demand elasticity σ_m to 2.56 as in Chari, Kehoe, and McGrattan (2000), while the scale factor χ is set to match the long-term ratio of the monetary base to output in the United States. The monopolistic markup factor is set to 20 percent, resulting from a demand elasticity for the differentiated products of $\epsilon = 6$. The average level of hours worked is set to one-third.

The steady-state debt-to-GDP ratio is assumed to be 50 percent, as the level to which debt would converge in the long run. The shock will drive up this debt by 65 percent. The steady-state government spending to GDP ratio is set to 20 percent. Since the short-term bond is only used to determine the stochastic discount factor, its actual quantity is zero. So the average maturity entirely depends on the properties of the long-term bond.¹⁴

The probability of the stochastic bond maturing is set to $\alpha = 0.0472$ to match the average maturity of US debt held by the public of 5.3 years, or 63 months, in US data. (see Section I). Recall that α also determines the fraction of debt maturing each quarter, and the fractions of currently outstanding debt that will mature in future periods. Figure 1 shows, as an example, how for this value of α the model gives a good representation of the declining fractions of debt outstanding for the United States.¹⁵

The parameters of the monetary policy rule assume the fairly standard values of $\phi_\pi = 1.5$ and $\phi_y = 0.5$, and a persistence parameter of $\rho_i = 0.75$. Conditional on the private agents' parameters, these values guarantee determinacy of the rational expectations equilibrium in models with balanced budget rules, as well as in models with sufficiently aggressive fiscal policy rules. In the baseline scenario, the inflation target is assumed to have a persistence of $\rho_\pi = 0.99$.

For the fiscal rule, we assume $\rho_\tau = 0.5$ and $\phi_\tau = 0.0037$, which yields determinate and nonexplosive equilibria in all our simulations.¹⁶ The tax response to deviations of debt from the steady-state sustainable level is very mild and thus gives the potentially strongest role for inflation to contribute to debt consolidation. Of course, a high tax responsiveness is possible, and could easily take care of the higher debt. We show its effects below, but this is the very scenario that political constraints will most likely make difficult to follow, and may be avoided by raising inflation

¹⁴ For the United States, this assumption is innocuous since the fraction of debt maturing α and the average maturity of debt $1/\alpha$ are close to US data. For other countries, allowing for non-infinitesimal quantities of the one-period bond would make it possible to calibrate at the same time any value for the average maturity of debt and the fraction of debt maturing.

¹⁵ The data used to compute the fractions of US debt maturing over the next ten years can be found in table 2, column 1 in Bohn (2011).

¹⁶ Even though ϕ_τ is small, the long-term response is $\phi_\tau/(1 - \rho_\tau)$, constituting passive fiscal policy, in the sense of Leeper (1991).

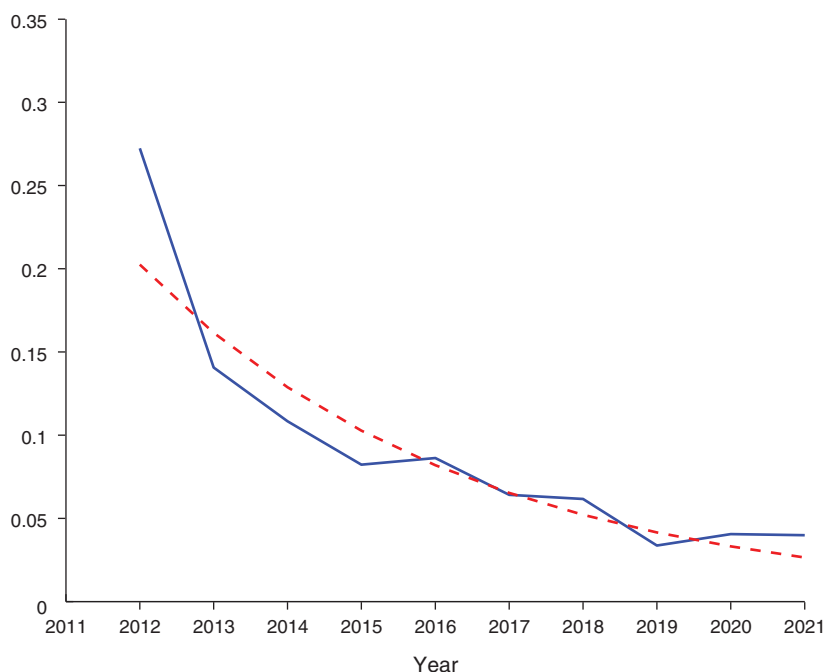


FIGURE 1. FRACTION OF DEBT MATURING

Notes: The solid line shows, for US debt outstanding in mid-2011, the fraction maturing over the next ten years. The dashed line depicts the corresponding fraction implied by the model.

instead.¹⁷ We describe the calibration of the shocks' standard deviations in the next sections along with the presentation of our different scenarios.

The state-space system describing the equilibrium dynamics of the full-information rational expectations solution of the linearized model can be found numerically, using standard methods. To describe the dynamics under imperfect information, that same state-space system can be used, but it needs to be augmented with the evolution of the inflation target that results from the application of the Kalman filter. Since the perception of the target determines its expected values in the future, choices of agents that depend on expectations in the current period are different than under full information.¹⁸

III. Analysis

In this section, we analyze the dynamic adjustment of public debt and other key variables with and without an exogenous change in the inflation target. The starting point of the simulations is an increase of public debt of the magnitude observed in the United States since the onset of the economic crisis of 2008 and 2009. We first

¹⁷Throughout, we assume that the model's approximation is far enough from a fiscal limit, where further tax increases would lead to falling tax revenue due to Laffer curve effects.

¹⁸Further details are given in the online Appendix.

TABLE 2—BASELINE CALIBRATION

Parameter	Value	Description
<i>Preferences</i>		
β	0.99	Time discount factor
σ_c	1.5	Intertemporal elasticity of substitution
σ_m	2.56	Inverse of the interest elasticity of money demand
χ	5.2×10^{-6}	Scale factor to utility of money balances, targets $m/Y = 0.07$
ϕ	2.00	Inverse of the Frisch of labor supply
φ	35.94	Scale factor to disutility of work, targets $h = 1/3$
<i>Bonds market</i>		
α	0.0472	Quarterly probability of maturing debt
<i>Firms</i>		
ϵ	6	Price markup of 20 percent
θ	0.75	One year price contracts
<i>Monetary policy</i>		
ρ_i	0.75	Interest rate smoothing parameter
ϕ_π	1.5	Response of interest rate to inflation
ϕ_y	0.5	Response of interest rate to output gap
<i>Fiscal policy</i>		
ρ_τ	0.5	Tax rate smoothing parameter
ϕ_τ	0.0037	Tax feedback to deviations of debt from steady-state

consider two scenarios for the response of tax rates to show how public debt would evolve absent change in the inflation target.¹⁹ Then we subject the inflation target to shocks of varying persistence.

A. A Debt Shock

In all the scenarios, debt is assumed to increase by about 65 percent from the current debt-to-GDP ratio. For the United States, this corresponds to the increase of debt from 54 percent of GDP in 2008 to the projected 89 percent in 2013. Figure 2 shows the subsequent evolution of real government debt. All variables responses are reported as percentage deviations from steady state, except inflation and the interest rates, which are expressed in annualized absolute deviations. The solid line depicts the dynamic response for a fiscal rule where the tax rate adjusts to higher debt just sufficiently to keep debt from exploding. Then real debt barely falls over the following 20 years. The associated tax rate increase is about 0.5 percentage points. Because of the higher stock of debt, of which each period a fraction $\alpha = 4.72$ percent becomes due, a correspondingly higher amount of new debt is issued each period.

The higher tax rate leads to a drop in after-tax *net* real wages for workers, reducing their labor supply and consequently output by about 0.2 percent below steady state. This distortionary effect lasts until debt returns to its long-run level. The inflation rate only initially slightly falls below the long-run target, following a short-lived contractionary effect of the tax rate on labor supply and thus the real wage, which

¹⁹The debt increase can be seen as a helicopter drop of government bonds, as in Leith and Wren-Lewis (2011).

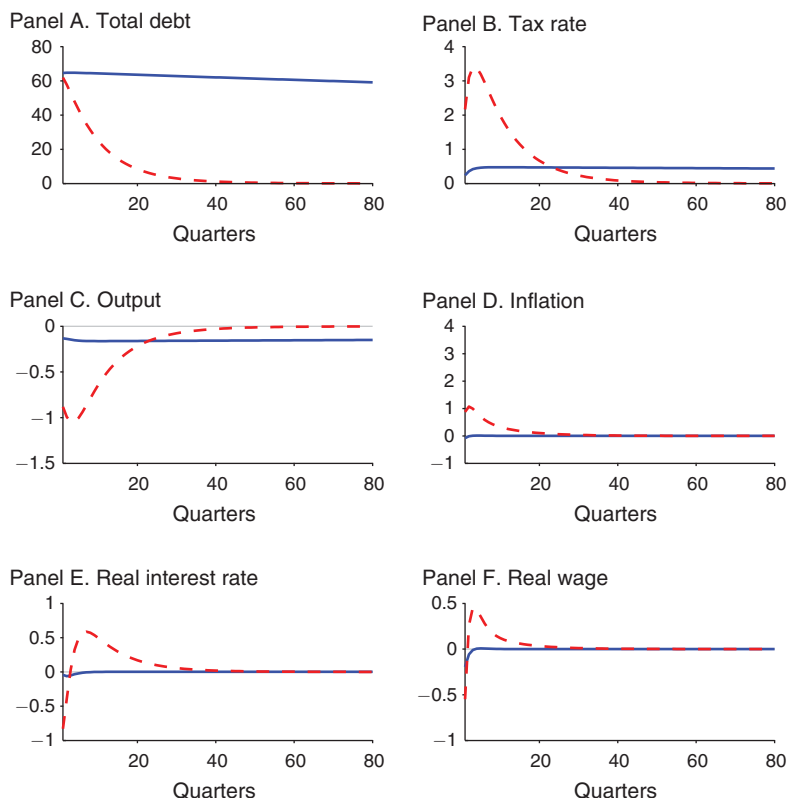


FIGURE 2. DEBT SHOCK

Notes: The impulse responses portray selected variables responses to the debt shock described in the text for two different scenarios. The solid line depicts the response of the economy under the baseline calibration while the dashed line illustrates the dynamics under a debt reducing tax policy.

reduces real marginal costs. However, after a few quarters, inflation, real wages, and the real interest rate return back to, or close to, the steady state.

Contrast this with a tax policy that reduces the additional debt within a short amount of time, depicted by the dashed line in Figure 2. This tax policy is represented in the fiscal authority's tax rule by a high coefficient on debt. The resulting tax increase is up to almost 4 percentage *points* above the steady-state tax rate, and leads to a much larger drop in output over several periods, almost 1.2 percent in the second quarter. At the same time, inflation rises by over 1 percentage point, which induces the policy interest rate, and also the real interest rate, to increase after a short drop. Finally, real gross wages rise because of the reduced labor supply after the tax increase.

B. Real Debt and Changes of the Inflation Target

We now turn to the question how much a change in the inflation target rather than raising taxes would contribute to a reduction in government debt. That is, we take the baseline change in debt to GDP of 65 percent, the fiscal rule that implies a

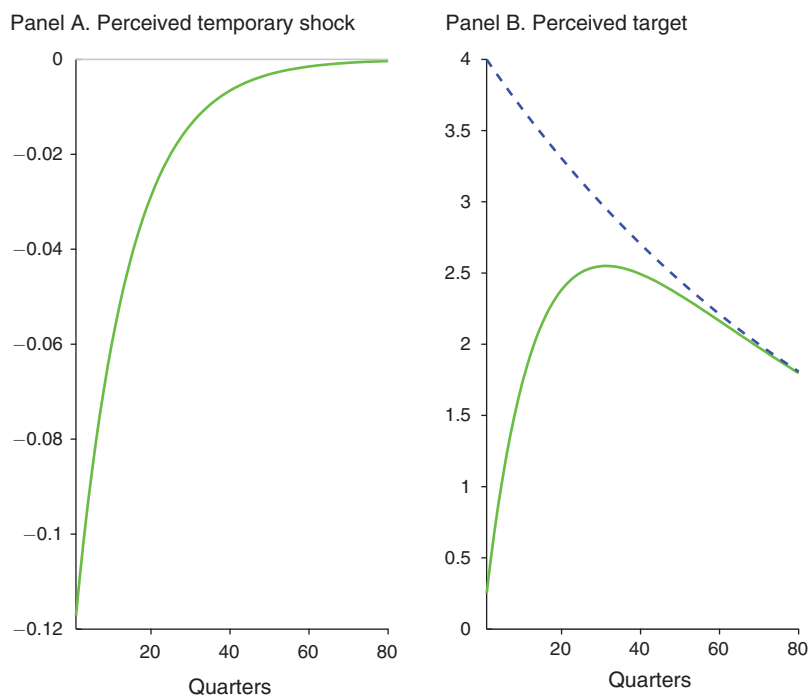


FIGURE 3. PERSISTENT TARGET SHOCK

Notes: The impulse responses portray selected variables responses to a persistent target shock for two different scenarios. The dashed line depicts the response of the economy under full information while the solid line illustrates the dynamics under learning.

minimal reaction of the tax rate, and simulate a persistent change in the annualized inflation target by 4 percentage points from 2 to 6 percent.

The persistence parameter of the target process is set to a high value of $\rho_\pi = 0.99$. Later, we consider less persistent changes.²⁰ Throughout, we compare here the evolution of the economy after changes in the inflation target both when it is perfectly observed and when the public cannot distinguish a change of the target from the transitory monetary policy shock. Recall that we take imperfect information to reflect a high credibility of a previously prevailing low inflation target, when the change in the target has not been communicated to the public. The degree of misperception depends crucially on how volatile the public perceives the inflation target to be, which in turn determines the speed of learning according to the Kalman filter. In the baseline calibration, the relative perceived volatilities of the target and the monetary policy shock are set such that perceived and actual inflation target coincide after 20 years.

Figure 3 shows the evolution of the actual and perceived inflation target after the one time increase of 4 percentage points. The target follows the process specified,

²⁰ Furthermore, for clarity, we assume that all the new debt is priced at the steady-state nominal interest rates, so that the shock to the inflation target occurs after the increase in debt. This avoids confounding effects of the target change with effects from the debt change as such.

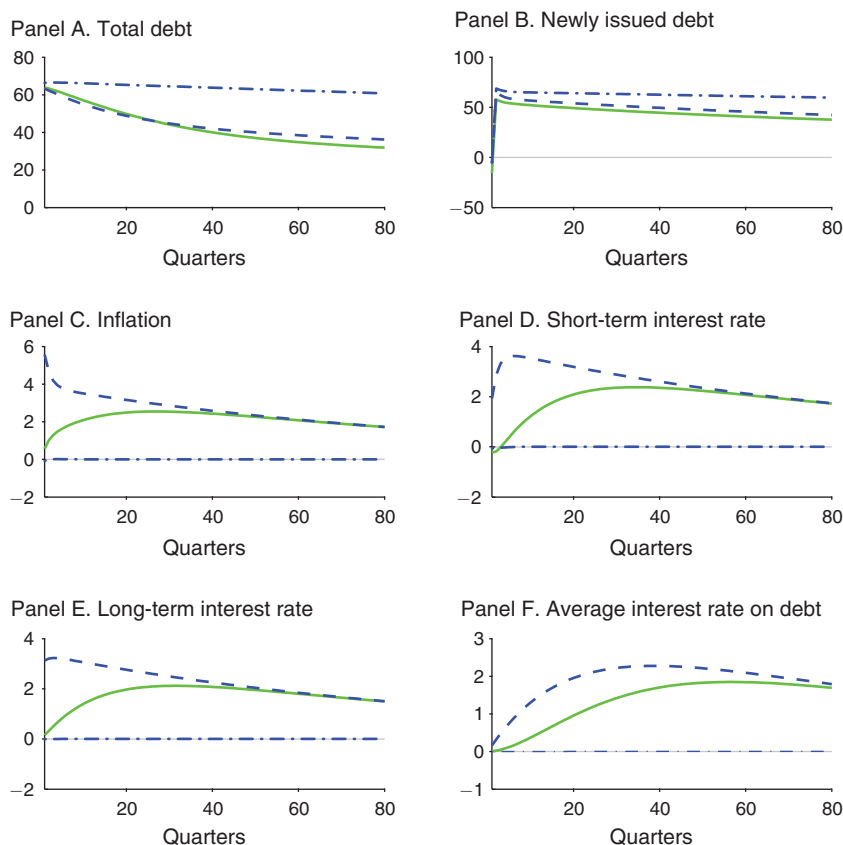


FIGURE 4. PERSISTENT TARGET SHOCK

Notes: The impulse responses portray selected variables responses to a persistent target shock for two different scenarios. The dashed line depicts the response of the economy under full information while the solid line illustrates the dynamics under learning. Finally, the dashed dotted line represents the benchmark constant target case.

as depicted by the dashed line in the right-hand side panel. In contrast, when agents only slowly learn about the changed inflation trend, it takes long for the difference between actual and perceived target to vanish. This is because, initially, agents assign a large fraction of the change in the nominal interest to the transitory shock, as can be seen clearly in the left panel.

The corresponding evolution of the economic variables of interest are shown in Figures 4 and 5. The former focuses on real government debt held by the public, the realized inflation rate, and the three measures of interest generated by the model. The latter figure shows the evolution of output, real interest and government spending on interest, the real wage, the marginal tax rate, and tax revenue.

Consider first the responses under full information about the inflation target in Figure 4, again depicted by the dashed lines. The addition to total debt that followed the debt shock falls over time, but at a decelerating rate. After 10 years, about 29 percent of this increase in total real debt has been inflated away. Recall that we show here only the additional debt above steady-state debt. Along with total debt, newly

issued debt follows a similar path in percentage deviations from its steady state, since a fraction $(1 - \alpha)$ of the now higher total debt has to be rolled over. The initial jump in new debt beyond what is needed to roll over is due to a substitution from money holdings to bonds, induced by a higher nominal interest rate that induces to a drop in money demand. The government budget constraint mandates a commensurate increase in bonds to offset the loss in seignorage.²¹

Actual inflation follows the same time path as the target, except for a short-lived initial cost push effect, which follows from the assumed inertial behavior of the short-term policy interest rate. Thus, the policy interest rate in the fourth panel does not immediately adjust to the higher target. Of course, after a few periods, the Fisher relationship has to hold, and the deviations of the short-term rate from steady state track that of inflation closely. This relationship is exploited in more detail below.

The behavior of the long-term interest rates on newly issued debt, $i_t^{L,n}$, shows the implications of the long-term bond introduced in this paper. To see this most clearly, linearize about the steady state the two Euler equations (3) and (4) for the short- and long-term bonds, to find that, up to first order: $i_t^{L,n} \approx \alpha^n i_t + (1 - \alpha^n) E_t i_{t+1}^{L,n}$, or

$$(21) \quad i_t^{L,n} \approx \alpha^n \sum_{s=0}^{\infty} (1 - \alpha^n)^s E_t i_{t+s},$$

where $\alpha^n \equiv (\alpha + i)/(1 + i)$. The long-term interest rate on newly issued debt is the weighted sum of all future expected short-term nominal interest rates, with declining weights $(1 - \alpha^n)^s$ as the time horizon s increases. This is borne out in the bottom left graph.

While the long-term interest rate is an average of future short rates, the average interest rate of outstanding debt is an average of all long-term rates set in the past. Thus, the average interest rate paid on debt can only sluggishly follow the evolution of the long-term interest rates on newly issued debt. This can be made explicit by linearizing equation (2), resulting in the recursion $i_t^L \approx \alpha^L i_t^{L,n} + (1 - \alpha^L) i_{t-1}^L$ or

$$(22) \quad i_t^L \approx \alpha^L \sum_{s=0}^{\infty} (1 - \alpha^L)^s i_{t-s}^{L,n},$$

where $\alpha^L \equiv (1 - (1 - \alpha)/\pi)$, which is close to α for the gross inflation rate π close to one. The average interest rate on outstanding debt is thus approximately a weighted average of all past long-term interest rates, with declining weights $(1 - \alpha^L)^s$, as the time s since issuance increases. This explains the slowly rising (dashed) line for the average interest rate on debt. Only after about 30 quarters it begins to fall again. Before turning to other variables of interest, we now discuss the adjustments under imperfect information.

The dynamics under imperfect information are depicted by the solid lines in Figure 4. The perceived inflation target differs from the actual target because agents assign a large fraction of observed interest rate changes to the monetary policy

²¹ Alternative calibrations of the money demand parameters imply slightly changed initial dynamics, but do not affect our main results.

shock, rather than the target. Recall the definition (18) of the signal, which is now filtered according to the Kalman filter. Consequently, actual inflation moves only slowly upward, since the perceived target $\tilde{E}_t \hat{\pi}_t^*$ enters firms' price setting, as given by the linearized Phillips curve for full indexation to the perceived target rate:

$$(23) \quad \hat{\pi}_t = \tilde{E}_t \hat{\pi}_t^* + \beta(E_t \hat{\pi}_{t+1} - \tilde{E}_t \hat{\pi}_{t+1}^*) + \kappa \widehat{mc}_t,$$

with κ a nonlinear function of the model's structural parameters. Thus, a high credibility of a previously established inflation target lowers the responsiveness of actual inflation to an inflationary change in the central bank's target.

The initial impact on real debt of the change on the inflation target under imperfect information differs only slightly from that under full information. The surprise effect of inflation on outstanding debt is smaller than before. This time, newly issued debt increases by less because of the slight drop in the nominal interest rate that leads to a small increase in seignorage revenue. The main difference to the full information case bears out over time, however, as agents underprice newly issued debt, because their inflation expectations are persistently lower than the actual inflation rates. This shows up in the correspondingly slow movements of the short- and long-term interest rates.

Figure 5 shows the movements of the remaining variables of interest. Output initially rises under both information scenarios because the inertial interest rate rule allows real interest rates to drop after the increase in inflation. Over time, output falls below steady state because of the distortionary effect of the higher tax rate. In proportion to the higher debt level, the public sector's spending on interest increases and follows the same dynamics. The behavior of labor tax revenue mirrors the dynamics of output and the tax rate.

C. Inspecting the Mechanism

We now turn to a qualitative analysis of how debt maturity and persistence of the inflation target determine the observed evolution of public debt. A few modest simplifications of the model allow us to give transparent analytical representations of the key mechanisms. First, we keep labor supply constant, which removes any feedback effects of taxes on output. Secondly, we ignore seignorage, for simplicity and because seignorage revenue is known to empirically play only a small role in government revenue dynamics for the ranges of inflation considered here. Finally, we assume perfectly flexible prices.

Under these assumptions, the equation for the evolution of debt combined with the real government budget constraint can be reduced to $\tau_t wN + b_t^L = g + (1 + i_{t-1}^L) b_{t-1}^L / \pi_t$. Then, with a slightly simplified fiscal rule $\tau_t = \tau + \phi_\tau (b_t^L / b^L - 1)$, we obtain

$$(24) \quad b_t^L = \frac{1}{1 + (\phi_\tau / b^L) wN} \left[g - (\tau - \phi_\tau) wN + (1 + i_{t-1}^L) \frac{b_{t-1}^L}{\pi_t} \right].$$

The evolution of debt essentially depends on the coefficient $(1 + (\phi_\tau / b^L) wN)^{-1}$ and the relative dynamics of i_{t-1}^L and π_t . The smaller the response coefficient in the

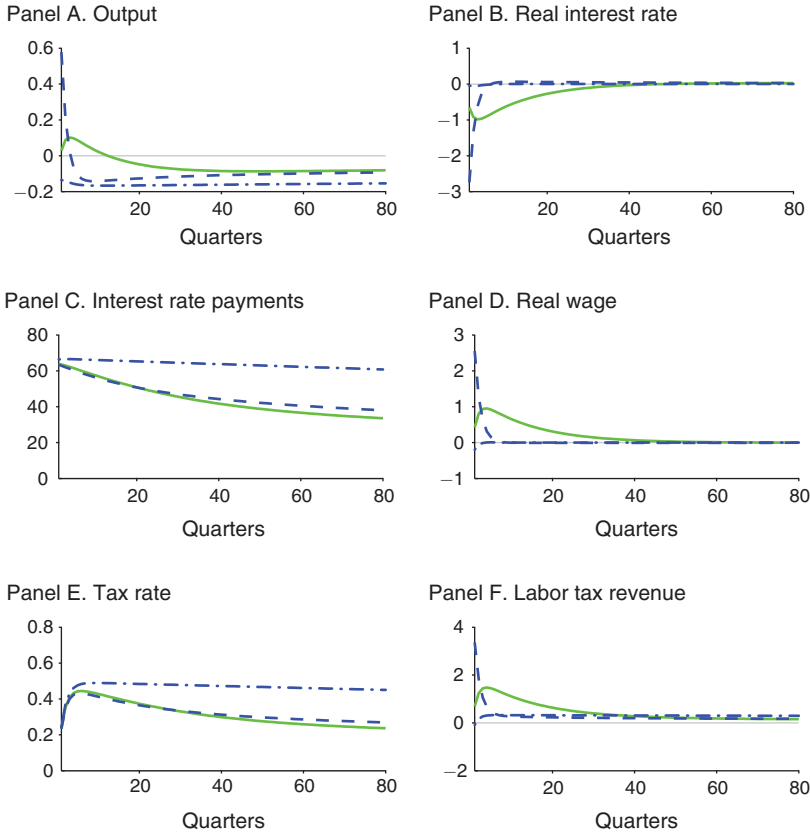


FIGURE 5. PERSISTENT TARGET SHOCK

Notes: The impulse responses portray selected variables responses to a persistent target shock for two different scenarios. The dashed line depicts the response of the economy under full information while the solid line illustrates the dynamics under learning. Finally, the dashed dotted line represents the benchmark constant target case.

tax rule, ϕ_τ the slower the adjustment of b_t^L will be to any variations of the right-hand side variables. However, as long as i_{t-1}^L and π_t do not act systematically to stabilize debt, ϕ_τ must be strictly positive and high enough to guarantee a non-explosive path of debt.²²

The future dynamics of i_t^L and π_t depend crucially on the expected path of inflation, which in turn depends on the expected path of the inflation target, π_t^* . This dependence can now be easily made explicit. The assumption of flexible prices implies that the monetary authority directly determines the inflation rate through its control over the short-term nominal interest rate. Since there are no movements in this simplified model's natural real rate of interest, $1/\beta - 1$, the evolution of the

²²In fact, for $\phi_\tau = 0$, the tax rate would be constant. Then, since in steady state, $g - \tau wN = (1 - (1+i)/\pi)b^L$, we can linearize equation (24) to get

$$\hat{b}_t^L = \frac{1+i^L}{\pi} \hat{b}_{t-1}^L + \frac{b^L}{\pi} \hat{i}_{t-1}^L - (1+i^L)b^L \hat{\pi}_t.$$

Since the real interest rate $(1+i)/\pi$ is larger than one in steady state, debt would be explosive up to first order.

short-term nominal interest rate is solely determined by the expected inflation rate. Setting $\rho_i = \phi_y = 0$ in the Taylor rule (14), and using the consumption-Euler equation (3), inflation can be easily found to follow²³

$$\hat{\pi}_t = \omega \hat{\pi}_t^* + \frac{1}{\phi_\pi} \eta_t.$$

where we have defined a scale factor $\omega \equiv (\phi_\pi - 1)/(\phi_\pi - \rho_\pi)$. The equation holds under full information, and shows that, after a change in the inflation target, future inflation can be expected to evolve directly proportional to the expected inflation target, since $E_t \hat{\pi}_{t+s}^* = \rho_\pi^s \hat{\pi}_t^*$ for $s \geq 0$, or

$$(25) \quad E_t \hat{\pi}_{t+1+s} = \omega \rho_\pi^{1+s} \hat{\pi}_t^*.$$

Under imperfect information, we have to substitute the target and monetary policy shocks by their respective perceptions at time t , i.e., $\tilde{E}_t \hat{\pi}_t^*$ and $\tilde{E}_t \eta_t$, which will be further used below. Then of course, actual and target inflation will not necessarily move closely together.

To determine the expected evolution of different nominal interest rates, we can make use of the two relationships (21) and (22) in combination with the consumption Euler equation (3) and the expected evolution of the inflation target, $E_t \hat{\pi}_{t+s}^* = \rho_\pi^s \hat{\pi}_t^*$. Then the Euler equation implies $E_t i_{t+s} - i = E_t \hat{\pi}_{t+1+s}$. Inserting (25) yields

$$E_t i_{t+s} - i = \omega \rho_\pi^{s+1} \hat{\pi}_t^*.$$

Using the approximation (21) for $i_t^{L,n}$, the interest rate on newly issued long-term debt can now be written as

$$(26) \quad i_t^{L,n} - i \approx \frac{\alpha^n \rho_\pi}{1 - (1 - \alpha^n) \rho_\pi} \omega \hat{\pi}_t^*.$$

Furthermore, inserting this into (22) delivers the average interest rate on outstanding debt as a function of the process for the inflation target

$$(27) \quad i_t^L - i \approx \frac{\alpha^n \rho_\pi}{1 - (1 - \alpha^n) \rho_\pi} \alpha^L \sum_{s=0}^{\infty} (1 - \alpha^L)^s \omega \hat{\pi}_{t-s}^*,$$

which is the second expression needed to characterize the evolution of government debt.²⁴

To gain some further intuition, consider first the factor in front of $\omega \hat{\pi}_t^*$ in equation (26). It reflects the relevant aspects of the forward-looking nature of long-term nominal

²³ To obtain this relationship, combine the simplified interest rate rule and the Euler equation and solve forward. Ruling out explosive paths gives the stated (unique) solution to the inflation rate.

²⁴ Recall that $\alpha^n = \frac{\alpha + i}{1 + i}$ and $\alpha^L = 1 - \frac{1 - \alpha}{\pi}$, which are close to α for low steady-state values of i and π .

interest rates: maturity of debt and expected evolution of inflation. When $\rho_\pi = 0$, that interest rate will be equal to the steady-state rate in all periods, $i_t^{L,n} = i$, since target inflation is independently and identically distributed about its steady state. In contrast, for an inflation target close to a random walk, $\rho_\pi \approx 1$, the interest rate follows the same process as the target. In fact, then also $\omega \approx 1$. Only for intermediate values of ρ_π does the maturity structure exert its influence on the long-term interest rate, which then on average compensates for future inflation rates. In contrast, for an economy with only short-term bonds, $\alpha = \alpha^n = 1$, as in the standard New Keynesian model, the nominal interest rate only needs to compensate for one period-ahead inflation. $i_t^{L,n} - i \approx \rho_\pi \omega \hat{\pi}_t^*$, which is of course the short-term nominal interest rate. For (27), the shorter the average maturity, i.e., α and thus α^L close to 1, the smaller the weights on past inflation rates will be. Then, again, the average rate tends to equal the long-term rate and the short-term rate as well. It is clear that under full information, unless α and ρ_π are at extreme values, the average long-term interest rate will be unable to compensate even for fully and correctly anticipated inflation that follows a change in the inflation target.

The equation for the evolution of real debt, (24), can now be rewritten in deviations from its steady-state level:

$$\hat{b}_t^L = \Phi \left[\frac{\alpha^n \rho_\pi}{1 - (1 - \alpha^n) \rho_\pi} \alpha^L \sum_{s=1}^{\infty} (1 - \alpha^L)^{s-1} \omega \hat{\pi}_{t-s}^* - \hat{\pi}_t + \hat{b}_{t-1}^L \right]$$

with $\Phi = 1 / (1 + (\phi_\pi / b^L) wN) \frac{1 + i^L}{\pi}$. This equation shows most directly the role of inflation persistence and average debt maturity α , and gives a simple characterization for the evolution of debt for different values for the parameters α and ρ_π . To further understand the role of the determinants of real debt dynamics, assume again that there is no long-term debt, i.e., $\alpha = 1$. Then $\alpha^n = \alpha^L = 1$, and debt follows

$$(28) \quad \hat{b}_t^L = \Phi [E_{t-1} \hat{\pi}_t - \hat{\pi}_t + \hat{b}_{t-1}^L]$$

since $E_{t-1} \hat{\pi}_t = \rho_\pi \hat{\pi}_{t-1}$. Under full information, a one-time increase in the inflation target would have an effect on real debt only in the period of the change, since future nominal rates adjust to compensate for the predictable path of inflation. That is, if $E_{t-1} \hat{\pi}_t = 0$, then a rise in inflation deflates real debt by $-\hat{\pi}_t$. But absent further shocks, there will be no expectational errors, since $E_t \hat{\pi}_{t+1} - \hat{\pi}_{t+1} = \omega(E_t \hat{\pi}_{t+1}^* - \hat{\pi}_{t+1}^*) = \omega(\rho_\pi \hat{\pi}_t^* - \rho_\pi \hat{\pi}_t^*) = 0$. In other words, without long-term debt, and under full information, only inflation surprises can affect the real value of the stock of outstanding debt. This is different under imperfect information.

Recall that the inflation rates $\hat{\pi}_t$ themselves are the outcome of realizations of the inflation target $\hat{\pi}_t^*$ or the beliefs $\tilde{E}_t \hat{\pi}_t^*$ under imperfect information. In the latter case, when agents slowly learn the true inflation target, they will make repeated expectational errors, even if all debt matures after one period. Thus real debt will be affected by inflation even when no further surprise shock to the target rate occurs. This explains the differences between the impulse responses under full information

and under imperfect information in the previous section. To be explicit, we can write the evolution of agents' optimal estimate of the target as ²⁵

$$(29) \quad \tilde{E}_t \hat{\pi}_{t+1}^* = \rho_\pi \tilde{E}_{t-1} \hat{\pi}_t^* + k'(\hat{\pi}_t^* - \tilde{E}_{t-1} \hat{\pi}_t^*).$$

The object of interest here are the future expectational errors for the inflation target $\tilde{E}_{t+s-1} \hat{\pi}_{t+s}^* - \hat{\pi}_{t+s}^*$ which, up to the factor ω , determine future expectational errors for inflation $E_{t+s-1} \hat{\pi}_{t+s} - \hat{\pi}_{t+s}$ in the equation for debt (28). It is easy to show that the expectational error for inflation must evolve according to

$$\tilde{E}_{t+s-1} \hat{\pi}_{t+s} - \hat{\pi}_{t+s} = (\rho_\pi - k')^s [\tilde{E}_{t-1} \hat{\pi}_t - \hat{\pi}_t]$$

after a one-time surprise increase in the target.²⁶ Under imperfect information, this expectational error only slowly declines as agents update their perception of the inflation target.

IV. The Sensitivity of Debt to Changing the Inflation Target

In this section, we return to the full model and explore the quantitative sensitivity of the results to variations in the parameters governing the persistence of the inflation target and the maturity structure of debt. It turns out that unless the deviation of the inflation target from its steady state is highly persistent, a significant effect on real public debt can only be achieved when the initial target change is very large. Also a high average maturity alone does not give a temporary but moderate change in inflation sufficient bite to have a large effect on debt. We also find that with mainly short-term debt, that the mispricing of newly issued debt arising from imperfect information amplifies the effect of higher inflation on debt.

To proceed, we need to decide on a metric that summarizes the effects of changing inflation targets on public debt. Recall Figure 4, where a large part of the debt reduction was achieved after 40 quarters, and the difference between full and imperfect information was noticeable. Therefore, we compute the relative percentage point difference between the real debt level under the changed inflation target and its level in the absence of that change, at a ten-year horizon. Formally, this is expressed by the following debt multiplier (DM hereafter) measure:

$$DM(h) = \left(\frac{\widehat{b}_{t+h}^{TS}}{\widehat{b}_{t+h}^L} - 1 \right) \times 100,$$

²⁵ Again, $k' = \frac{\rho_\pi(1 - \phi_\pi)\mathcal{P}'}{(1 - \phi_\pi)^2\mathcal{P}' + \sigma^2}$ is the Kalman gain parameter, where $\mathcal{P}'$ solves the equation $\mathcal{P}'^2 + [(1 - \rho_\pi^2)\sigma^2/(1 - \phi_\pi)^2 - \sigma_\pi^2]\mathcal{P}' - (\sigma_\pi\sigma/(1 - \phi_\pi))^2 = 0$.

²⁶ The future evolution of the perception after a one-time target shock is

$$\tilde{E}_{t+s-1} \hat{\pi}_{t+s}^* = \rho_\pi^s \hat{\pi}_t^* + (\rho_\pi - k')^s [\tilde{E}_{t-1} \hat{\pi}_t^* - \hat{\pi}_t^*],$$

while the actual evolution of the target after the shock at time t is $\hat{\pi}_{t+s}^* = \rho_\pi^s \hat{\pi}_t^*$. For persistent target shocks and noisy signals, ρ_π close to one and k' small, $(\rho_\pi - k')^s$ will decline monotonically in s .

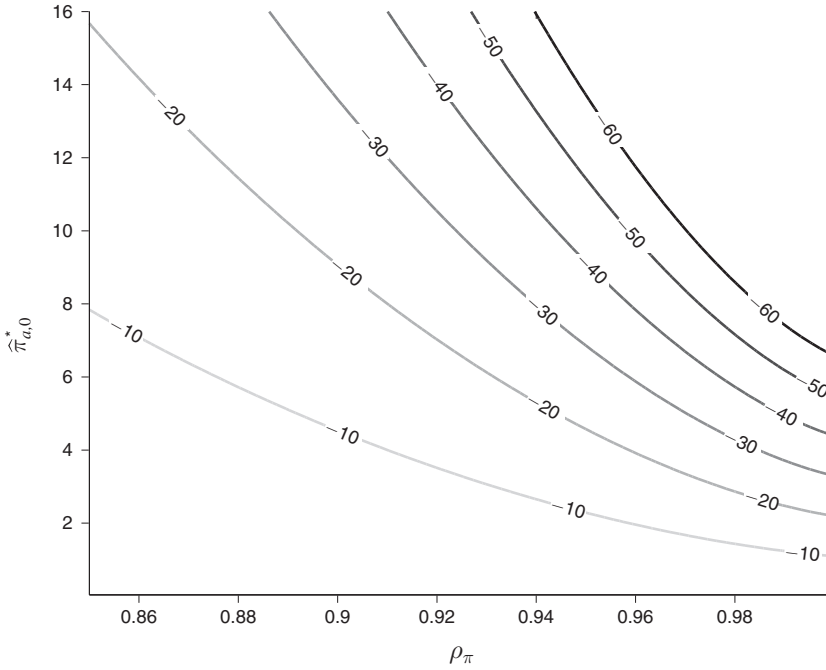


FIGURE 6. INFLATION TARGET PROCESS AND SHOCK SIZE

Notes: Difference in debt reduction $DM(40)$ contour plot as a function of the parameters $\hat{\pi}_{a,0}^*$ and ρ_π . Keeping the other parameters fixed, these parameters are varied in the reported range.

with $h = 40$ quarters, and where $\hat{b}_{t+h}^{L,TS}$ and $\hat{b}_{t+h}^L$ are, respectively, the percent deviations from steady state of the levels of real debt after the target shock that occurs at time t , and the level of debt under no target shock. Thus, the measure basically compares the difference between the dashed-dotted line and the solid or dashed lines in Figure 4.

A. The Process of the Inflation Target

In the baseline scenario, the annual inflation target was increased by 4 percentage points and very slowly reverted to the low-inflation, steady-state target. This essentially amounts to a permanent departure from a previously communicated monetary policy strategy. However, a decision maker may be contemplating a policy where inflation increases only temporarily, which corresponds to the proposals made by Rogoff (2008, 2010) and others. We therefore analyze here the role of the persistence and size of the innovation to the inflation target, mainly focusing on the full information case.

Figure 6 depicts those combinations of ρ_π and $\hat{\pi}_{a,0}^*$ that yield a given relative reduction $DM(40)$ in real public debt after 10 years (40 quarters). Here, $\hat{\pi}_{a,0}^* = 4 \times \hat{\pi}_0^*$ denotes the annualized increase of the target at time $t = 0$. The

baseline scenario is given by $\rho_\pi = 0.99$ and $\hat{\pi}_{a,0}^* = 4$, which results in a drop of about 30 percent in the real value of debt. Larger initial target increases for a given persistence yield proportionately stronger effects on real debt. A smaller increase, of only 2 percentage points, to the 4 percent inflation target proposed by Blanchard, Dell'Ariccia, and Mauro (2010) would therefore only reduce real debt by 15 percent after 10 years. In contrast, when the persistence is decreased, the effect of a given target change declines more than proportionately. Thus, for a persistence of, say, $\rho_\pi = 0.95$, a 30 percent reduction in real debt would only be achieved if the target instead jumps by $\hat{\pi}_{a,0}^* = 8$ annualized percentage points above the steady-state target.²⁷

In general, if a policymaker wanted to achieve a particular reduction in real debt by raising inflation temporarily, as advocated by Rogoff and others, the effect would be moderate unless the initial jump in the target is very large. An obvious reason is that there is less time for inflation to affect outstanding debt. But also, actual inflation increases by somewhat less than the target itself, due to the scale factor ω , which is declining with ρ_π . Finally, in the model, interest rates on newly issued debt are set to compensate higher inflation on average. Thus, even though inflation may have fallen, some of the debt issued after the target change will still require a higher interest service. As can be gathered from the impulse responses in Figure 4, the corresponding effects on real debt would only be slightly higher under imperfect information.

B. Debt Maturity and Credibility

The importance of the maturity structure for the susceptibility of real public debt to a higher inflation target is illustrated in the left panel of Figure 7. The change in the target is the 4 percentage point increase considered before, and the vertical line indicates the baseline value $\alpha = 0.0472$ for the fraction of debt that matures each period. Recall that $1/\alpha$ is then the average expected maturity of debt, in this case almost five and a half years. For the baseline scenario, the intersections of the vertical line with the dashed and solid curves show the debt reductions after 10 years under full and imperfect information, respectively. While under full information, the debt reduction is a little under 30 percent, under learning it is 34 percent.

Higher values of α imply lower average maturities of debt, so that the effect on debt of an increase in the inflation target shrink relative to the baseline scenario. In the extreme case with one-period bonds only ($\alpha = 1$), the reduction in real debt after a fully perceived increase in the target falls to about 10 percent.²⁸ By contrast, if a change in inflation is only slowly perceived to be due to a change in the target, then interest rates on rolled-over debt are repeatedly set too low to compensate for inflation, and real debt can be deflated by almost 25 percent after 10 years. Thus,

²⁷ For $\rho_\pi = 0.95$, the average annual target inflation rate over ten years would be 3.48 percentage points above the steady-state target after the 8 percentage point shock. For $\rho_\pi = 0.99$, the inflation target would in this case be on average 6.62 percentage points above steady state.

²⁸ In fact, when $\alpha = 1$, and the target change is fully-perceived, the drop in real debt is mainly achieved by the initial surprise jump in inflation that deflates the existing debt before it is rolled over at a higher nominal interest rate.

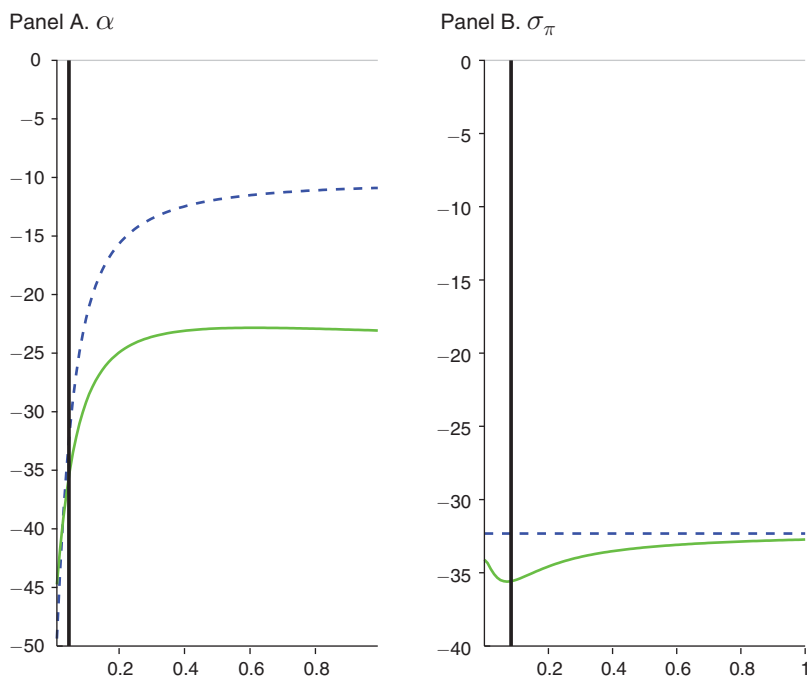


FIGURE 7. AVERAGE MATURITY AND CREDIBILITY

Notes: Difference in debt reduction $DM(40)$ as a function of the parameter on top of each panel. Keeping the other parameters fixed, that parameter is varied in the reported range. The vertical solid bar indicates its baseline value. The dashed line and the solid line, respectively, depict the full information and the learning case.

the shorter the average maturity of public debt, the higher is the role of more firmly anchored past inflation expectations on the sensitivity of real debt to higher actual inflation.

A higher maturity implied by values of α below 0.0472 further facilitates inflating away the debt. In a “British” scenario, with an average maturity of almost 14 1/2 years ($\alpha = 0.0174$), much higher than the close to 5 1/2 years for the United States. In that case, after 10 years, inflation would reduce the additional debt by more than 40 percent, irrespective on whether there is full information or not. The difference between full information and imperfect information remains relatively small, as in the baseline. Thus, for realistic maturity structures, the credibility of a previously established inflation target does not strongly affect the degree to which higher inflation can reduce the real value of government debt.

The right panel of Figure 7 shows the role of the *perceived* volatility σ_π of the inflation target on the reduction of debt, after a given target change of four percentage points. The perceived volatility of the inflation target relative to the perceived volatility of the monetary policy shock determines the gain in the Kalman filter, and thus the speed of learning. In other words, we illustrate here for the case of imperfect information how much the effect of inflation on debt is changing when agents interpret a change in the signal ε_t^π more or less strongly as a change in the inflation target. Again, the vertical line in the graph depicts the baseline case, which corresponds to

the dynamics of learning as shown in Figure 4. The more of a change in the signal ε_t^π is assigned to a change in the target, that is, the higher σ_π , the lower is the benefit from not fully communicating the target change. Conversely, for a the lower σ_π , agents believe that a change in the target is less likely, and therefore stick to their previous inflation expectations. Then the effect on real debt is much larger, since nominal interest rates compensate too little for subsequently higher inflation.²⁹

C. The Zero Lower Bound and Debt

The preceding sections focused on the effects of a changed inflation target per se on the real value of public debt. For concreteness, a debt shock was assumed that increases real debt from an initial level by an amount comparable to the large increases after the economic crisis of 2008 and 2009. But the proposals to increase inflation to reduce the debt burden were made in the specific context of the severe contraction that followed the collapse of the US housing market and the resulting strains on the financial sector. Even though monetary policy responded by setting short-term interest rates close to their lower bound of zero, inflation remained very low by historical standards and unemployment recovered only very slowly. Measures of quantitative easing may have helped to avoid a protracted slump and a fall into deflation, but they have been unsuccessful in generating a sustained recovery. In such an environment, with interest rates bounded from below, the ability to raise inflation may be limited. This section broadens the analysis by considering such a scenario.

To generate in dynamic general equilibrium models a contraction of output and inflation that is large enough to force the nominal interest against the zero lower-bound, we follow the literature by introducing an inter-temporal preference shock.³⁰ Formally, the discount factor is assumed to be time-varying and to be following an AR(1) process. Consequently, the subjective discount factor β of households is now defined as $\beta_t \equiv \beta \delta_t$, with $\log(\delta_t) = \rho_\delta \log(\delta_{t-1}) + \eta_t^\delta$, and $0 < \rho_\delta < 1$, and with η_t^δ an independently and identically distributed disturbance.³¹ We set the persistence parameter to $\rho_\delta = 0.9$. Following Erceg and Lindé (2014), we increase the degree of price stickiness to $\theta = 0.9$. This leads to a lower response coefficient on marginal costs in the Phillips curve, from 0.0858 to 0.0121, thus weakening an otherwise overly strong effect of output on inflation. This is crucial for limiting the fall of inflation to a more plausible magnitude.³² Furthermore, since the crisis led to an unprecedentedly swift reduction in the policy interest rate, we set the interest rate smoothing parameter in the Taylor rule to $\rho_i = 0$. The size of the shock is set to result in eight quarters during which the zero lower bound on the short-term nominal interest rate binds.

²⁹ Melecký, Rodríguez Palenzuela, and Söderström (2009) estimate the volatilities of the inflation target and the policy shock, and interpret a higher perceived volatility as an overestimate by households.

³⁰ See, among many others on zero lower bound episodes, Christiano, Eichenbaum, and Rebelo (2011) and Woodford (2011).

³¹ The household intertemporal utility now reads $E_0 \sum_{t=0}^{\infty} \beta^t \prod_{s=0}^{t-1} \delta_s \left(\frac{C_t^{1-\sigma_c}}{1-\sigma_c} + \chi \frac{(M_t/P_t)^{1-\sigma_m}}{1-\sigma_m} - \varphi \frac{N_t^{1+\phi}}{1+\phi} \right)$.

³² The lower value of the response coefficient lies well within the range from 0.005 to 0.437 of the available DSGE-based empirical estimates reported in Schorfheide (2008).

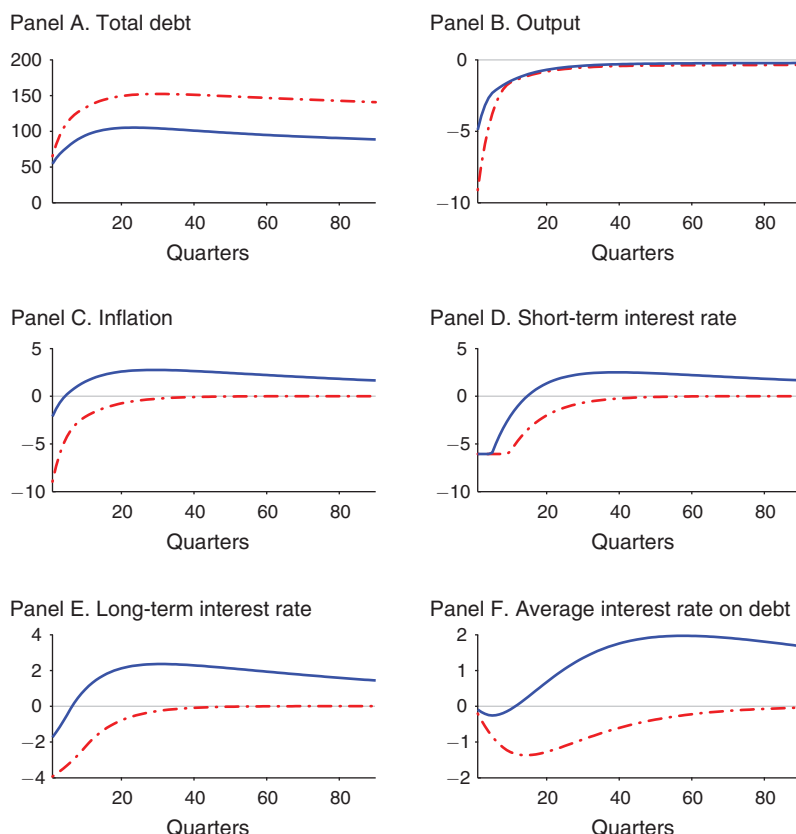


FIGURE 8. PERSISTENT TARGET SHOCK AND THE ZERO LOWER BOUND

Notes: The solid line depicts the impulse responses of selected variables responses to a persistent target shock. The dashed dotted line represents the benchmark constant target case.

Figure 8 shows the adjustment of selected variables after a positive discount factor shock (the thin, dash-dotted line) when leaving the inflation target unchanged.³³ All the variables are reported as percentage deviations from steady state, except inflation and the interest rates, which are expressed in annualized absolute deviations. On impact, both inflation and output drop by 7.2 and almost 8 percent, respectively. These are large magnitudes, but consistent with those reported in other studies. Even though the initial increase in total debt is 65 percent, as after the debt shock, the resulting dynamics of debt are substantially different. First, the contraction in aggregate output leads to the strong and persistent drop in tax revenue. Second, the resulting negative inflation rate substantially increases the real value of outstanding debt. Both lead to the initial jump and the continued increase in debt. While the short-term nominal interest rate initially stays at zero, long-term and average interest rates on debt show a more muted response, consistent with the pattern seen in Figure 4.

³³ The rational expectation equilibrium dynamics under the zero lower bound constraint are derived numerically combining the Dynare toolbox (Adjemian et al. 2011) and the algorithm developed in Holden and Paetz (2012).

Under full information, a simultaneous increase in the inflation target and the discount factor has the effects depicted by the solid line in Figure 8. As there is no uncertainty regarding the true value of the inflation target, agents fully understand the change in monetary policy and price setters lower their prices by much less than when only the discount factor falls. As a result, the real value of public debt increases by much less than without the target change. Since also the real interest rate now rises by less, the output contraction and drop in tax revenue turn out to be milder. The drop in the nominal interest rate is more short-lived, leading to an escape from the zero lower bound already after five quarters. After ten years, public debt is about 32 percent lower than without the target shock, close to the results after the debt shock.

A change in the inflation target appears to be an effective tool not only to mitigate an increase in real public debt, but also to help the economy exiting from the zero lower bound and reduce the contraction in output. However, the responses are driven solely by expectations of future inflation, a mechanism familiar from the discussions of central bank forward guidance.³⁴ Typically, forward guidance amounts to announcements of lower than necessary nominal interest rates after the economy has exited the zero lower bound. Through expected higher inflation also current inflation would be raised, with a resulting fall in real interest rates and rising economic activity. The problem with this mechanism is that it relies on announcements of future policies that may be time-inconsistent, as policymakers have no incentive to keep inflation high once the economy has exited the lower bound. Thus, the credibility of a promised high inflation rate may be low.

Considering an extreme case of imperfect information is instructive for understanding what happens when the above expectations channel is weak. For this, assume that there is no information about an intended target change. In line with our earlier interpretation of credibility, this amounts to assuming full credibility of a previous commitment of the previously established inflation target, or conversely, a complete lack of credibility of an announced target change. Then the only channel through which inflation can be increased (or deflation reduced) is via contemporaneous increases in real marginal costs. But under the zero lower bound constraint, nominal interest rates cannot fall to generate the rise in aggregate demand that would push up marginal costs. Thus, until the point where the nominal interest rate is no longer bound at zero, there can be no effect of a change in the inflation target. After that, a higher target would lead the central bank to keep the nominal interest rate at zero longer, which only then can lead to higher inflation as marginal costs rise, and thus only then to a decline in real public debt.³⁵

V. Conclusion and Discussion

This paper investigated to what extent and under which conditions a higher inflation target reduces the real value of public debt. We took the proposal by Rogoff

³⁴ See Levin et al. (2010) and Eggertsson and Woodford (2003).

³⁵ While the case of no information is intuitive, a formal analysis of intermediate cases is beyond the scope of this paper.

(2010) as a starting point, who suggested that a temporary, e.g., two- to three-year, increase of US inflation of about four percentage points would significantly alleviate that country's public sector's balance sheet. Our main finding is that only a permanent increase of the targeted inflation rate from 2 to 6 percent would result in a sizable reduction of about 30 percent of the additional government debt accrued since the crisis that began in 2008. By contrast, the proposed temporary change in inflation has a substantially smaller effect, of at best 10 percent of debt after 10 years. Furthermore, for realistic average maturities of public debt, the credibility of a previously established inflation target does not strongly affect the extent to which pushing up inflation reduces real government debt. That is, unless debt were mainly of short maturity, not much would be gained from misleading the public by attempting an inflationary policy without revealing it. In the presence of the zero lower bound, the ability to use forward guidance by credibly committing to future inflation would have positive effects on current inflation.

In our analysis, both the process that the inflation target is allowed to follow and the maturity structure are given. In the remainder of this section, we discuss the likely consequences of relaxing these assumptions, and will argue that this would further reduce the effect of inflation on the real value of public debt.

The long-term inflation expectations in the model are largely driven by the public's beliefs about the level of the inflation target. At the same time, the persistence and volatility of the target and the volatility of the monetary policy shock are assumed given and known, which allowed agents to forecast the path of inflation, given that perceived level of the target. However, also the volatilities could be potentially time-varying and subject to an analogous information imperfection. An observed higher inflation then may not only induce updates of the implied inflation target but also of the volatility of the target. This would affect the premium that investors demand for nominal risk, putting additional pressure on the government budget through higher interest rates, possibly even offsetting the intended reduction in real public debt.

Ultimately, underlying the public's beliefs about an inflation target (and other aspects of monetary policy) must be beliefs about the commitment of the central bank, as well as beliefs about whether the government will respect that commitment. In technical terms, the public's assessment of inflation risks may be the result of an inference about the type of policymaker in charge. Those beliefs are likely to be shaped by predictions regarding the government's willingness and ability to use other means of budget consolidation, which in the end depends on how the political system resolves the distributional trade-off that arises between these options. The public also needs to form beliefs about how a conflict between the monetary and fiscal authorities over monetary policy is resolved. All these aspects affect the credibility of policy and might influence inflation expectations.³⁶

Also the maturity structure is in actuality endogenous and a choice of the government, and thus may carry information about the inflationary intentions of the

³⁶ Arguably, these issues become even more complex when considering a monetary union, such as the Euro-Area, as a conflict between monetary authority and many fiscal authorities, and diverging interests among those fiscal actors have to be factored in.

government: on the one hand, if it contemplates inflating away debt it may want to lengthen the average time to maturity of its debt, to increase the potential effects of inflation later on. On the other hand, the market anticipating this would price in higher inflation expectations. Thus, by leading to higher long-term interest rates early on, the government's intentions may be countervailed. In fact, a government concerned about its reputation may then in fact want to do just the opposite: lowering maturity signals an absence of intentions to raise inflation or to put pressure on the central bank, but rather shows a commitment to stick to the inflation target.

Our analysis deliberately remains silent on the details of these deeper issues of credibility, commitment, and political economy, since also a more complex analysis is likely to find an even weaker role for inflation to change the real value of public debt. Implicitly, we are thus contemplating a best-case scenario, identifying an upper bound on what a government may be able to achieve by raising the inflation target, under the most favorable circumstances.³⁷

Finally, we focused on a closed economy. One factor which may work toward increasing the effectiveness of inflation is the fraction of debt held by foreigners. As long as debt is denominated in a country's own currency, a devaluation following higher inflation may reduce the real debt burden further, since fewer real resources are transferred to foreigners (See Aizenman and Marion 2011, for such a scenario). However, at the current juncture after the Great Recession of 2008 and 2009, where many countries face similar problem, it is questionable whether a small open economy is a suitable benchmark. Rather, other countries may choose to inflate at the same time, leaving the exchange rate unchanged, so that the results for the analysis of a closed economy are likely to generalize.

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³⁷ There is a growing literature addressing some of the interactions that may arise. See, for example, the work described in Aguiar et al. (2014).

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